

Rodolfo G. Lotte, PhD

SENIOR MACHINE LEARNING ENGINEER | COMPUTER VISION | AI SOLUTION ARCHITECT

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Professional Summary

Senior Machine Learning and Computer Vision Engineer with 12+ years of experience designing, training, and deploying production-grade ML systems across remote sensing, industrial inspection, environmental monitoring, LLM, and geospatial analytics. Certified AWS Solution Architect - Associate. PhD in Remote Sensing with deep expertise in deep learning for imagery (CNNs, segmentation, detection), large-scale data pipelines, and AWS-based architectures. Proven track record delivering national-scale solutions, leading multidisciplinary teams, and translating research into robust, scalable AI products.

Core Skills

Agentic AI & Vision

- **Agentic Workflows & Orchestration:** Designing multi-agent systems, autonomous tools, and stateful graphs using **LangGraph** and open-source **MCP** (Model Context Protocol) servers.
- **Production AI & Computer Vision:** Production-grade Semantic Segmentation, Object Detection, and **Super-Resolution** pipelines optimized via **PyTorch** and **TensorRT**.
- **Advanced GenAI & Evaluation:** High-performance RAG architectures, LLM fine-tuning execution, and automated LLM evaluation/guardrail frameworks for enterprise reliability.

Backend & MLOps

- **High-Performance Backend Engineering:** Designing scalable microservices, asynchronous architectures, and production-grade APIs using **FastAPI** and **Pydantic**.
- **CI/CD & Automated MLOps:** Continuous integration and deployment pipelines for ML workflows, automated retraining loops, and robust experiment/dataset tracking (**MLflow**, **DVC**).
- **Inference Infrastructure:** Engineering optimized pipelines for high-throughput batch inference and real-time streaming systems.

Cloud & Architecture

- **Cloud-Native ML Architectures:** Secure, scalable infrastructure design and deployment on **AWS** (ECS, Lambda, Bedrock, SageMaker, DynamoDB, IAM).
- **Scalable System Design:** Solution design for distributed, high-volume data streams and parallelized processing workloads.

Geospatial Engineering

- **Remote Sensing Analytics:** Large-scale processing of multi-spectral satellite, aerial, SAR, and LiDAR datasets for urban planning, 3D reconstruction, cadastre updates, and environmental monitoring.
- **Programmatic GIS:** Advanced spatial data manipulation, indexing, and pipeline development using **PostGIS**, **GDAL/OGR**, **Rasterio**, and **Fiona**.
- **3D Visualization Platforms:** System architecture design for high-fidelity, GPU-accelerated WebGIS platforms and spatial data engines.

Core Programming

- **Production Languages:** Advanced **Python** (asyncio, functional programming principles, optimized data structures), **SQL**, and **C++**.
- **Protocol & API Design:** Building custom enterprise data integrations using production RESTful APIs and standardized agentic communication protocols.

Education

Data Science Academy - DSA (Faculdade Vincit)

POSTGRADUATE IN MACHINE LEARNING ENGINEER 4.0

São Paulo, Brazil

Jan. 2025 - Jan. 2026

- Postgraduate Module in Machine Learning Engineering
- Development and Deployment of Machine Learning Models
- CI/CD Pipelines for Machine Learning and AI Operations
- Software Engineering for Machine Learning

National Institute for Space Research (INPE)

PHD IN REMOTE SENSING

SJC, São Paulo, Brazil

Mar. 2014 - Aug. 2018

- Topic: 3-dimensional (3D) urban mapping: A study of detection and reconstruction of building's facade through Structure-from-Motion (SfM) and Convolutional Neural Network (CNN)

Institute für Photogrammetrie (IfP)

PHD EXTENSION

Stuttgart, Germany

Jul. 2017 - Jan. 2018

- Topic: 3-dimensional (3D) urban mapping: A study of detection and reconstruction of building's facade through Structure-from-Motion (SfM) and Convolutional Neural Network (CNN)

National Institute for Space Research (INPE)

MSC. IN COMPUTER SCIENCE AND ENGINEERING

SJC, São Paulo, Brazil

Mar. 2010 - Dec. 2012

- Topic: Road extraction in airborne SAR images: An Active Contour Model approach using semiautomatic seeding

University of Franca (UNIFRAN)

B.S. IN SYSTEM ANALYSIS

Franca, São Paulo, Brazil

Mar. 2006 - Dec. 2009

- Topic: Conexionist knowledge in a image-based search engine

Experience & Projects

Imagem Geossistemas — Esri Official Distributor

BUSINESS AI ARCHITECT

SJC, Brazil

Jul 2026 – Present [full-time]

- Architected scalable, cloud-native GIS solutions leveraging the ESRI ecosystem and robust cloud infrastructure (AWS/Azure) to address complex public administration and urban management challenges.
- Designed and orchestrated agentic AI workflows and automated machine learning pipelines, integrating disparate public data services into cohesive decision-support systems for government clients.
- Led the technical integration of advanced remote sensing capabilities—such as semantic segmentation and super-resolution—into geospatial architectures to modernize public cadastre and territory monitoring.

Deloitte — Strategy, Data & AI

SENIOR DATA SCIENTIST / ML ENGINEER

São Paulo, Brazil

Jun 2023 – May 2026 [full-time]

- Designed and deployed production-grade ML solutions for large industrial and energy clients, spanning computer vision, predictive modeling, and data validation pipelines.
- Architected AWS-based ML workflows enabling continuous training, inference, and monitoring across multiple applications.
- Led technical design of computer vision systems for industrial inspection and operational monitoring.
- Collaborated with cross-functional teams (engineering, business, operations) to translate complex requirements into scalable, deployable AI solutions.
- **Key Clients**
 - **Petrobras — Well Drilling Engineering:** Built ML-driven pipelines to support engineering decision-making and large-scale analysis of operational data; designed scalable data organization strategies and automated ML workflows to enable continuous model updates.
 - **Petrobras — Computer Vision:** Designed and implemented computer vision solutions for operational monitoring, including object detection pipelines, cloud-based processing, and end-to-end ML lifecycle management.

Bioverse Labs

MACHINE LEARNING AND REMOTE SENSING LEAD

São Paulo, Brazil

Jul. 2020 - May. 2024 [part-time]

- Led development of deep learning systems for tree species detection using very high-resolution satellite imagery.
- Designed modular pipelines for image preprocessing, training, inference, and evaluation, enabling rapid experimentation and reproducibility.
- Achieved high-accuracy species classification across heterogeneous environments by leveraging spectral, spatial, and contextual features.
- Represented the team in the XPRIZE Rainforest competition, contributing to large-scale ecological monitoring solutions.

Instituto de Manejo e Certificação Florestal

COMPUTER VISION AND REMOTE SENSING SPECIALIST

Piracicaba, São Paulo, Brazil

Oct. 2022 - May. 2023 [part-time]

- Contributed to RAMI (Radar Mining Monitoring), a platform for detecting illegal mining in the Amazon.
- Developed and validated deep learning models for mining detection using high-resolution open satellite data (NICFI).
- Supported operational monitoring across large geographic areas in Pará, improving detection reliability and validation workflows.

Santiago & Cintra Consultoria, SCON

COMPUTER VISION AND REMOTE SENSING SPECIALIST

São Paulo, Brazil

Dec. 2020 - May. 2023 [full-time]

- Designed AI-driven methodologies for public security, environmental monitoring, and urban cadastre
- Built automated pipelines for image selection, enhancement, classification, and validation.
- Supported federal institutions in deforestation monitoring and land-use analysis.

Programa das Nações Unidas para o Desenvolvimento - ONU

FULL STACK DEVELOPER - IC CONTRACT

Brasília, Brazil

Mai. 2021 - Jun. 2022 [part-time]

- Keycloak user migration: project to merge Keycloak instances via Keycloak API REST, done in Python language.
- Keycloak integration: integration of Codex and Sinapses AI applications with particular SSO authentication services - unification and external auth merges.
- Support to issue legacy: support to issues over Codex and Sinapses AI applications in the PDPJ project, under Conselho Nacional de Justiça (CNJ) division.

Amazon deforestation monitoring program, PRODES-AMZ

SENIOR DEVELOPEMENT - AUTONOMOUS PROCESS IN REMOTE SENSING

INPE, SJC, São Paulo, Brazil

Jan. 2020 - Dec. 2020 [full-time]

- Developed deep learning models to classify and detect recent deforestation events using aerial and satellite imagery.
- Implemented cloud-based processing pipelines on AWS, enabling scalable image ingestion and classification.
- Automated end-to-end workflows from data acquisition to inference, supporting national-level environmental monitoring.

Geopixel Soluções em Geotecnologias e TI

TECHNICAL LEAD IN REMOTE SENSING

Parque Tecnológico, SJC, São Paulo, Brazil

Jun. 2019 - Dec. 2019 [full-time]

- Led teams of 8–10 analysts in large-scale land-use mapping projects.
- Managed delivery of statewide datasets, including: 199,000 km² mapped into 13 land-use classes
- Provided technical leadership, QA oversight, and GIS/ML guidance across projects.

Brazilian Space Weather Monitor Program, EMBRACE

SOFTWARE ENGINEER / ASTROPHYSICS RESEARCH

INPE, SJC, São Paulo, Brazil

2011 - 2019 [full-time]

Brazilian Space Weather Monitoring Program - [Read More](#)

- Designed and maintained data-intensive systems for monitoring ionospheric and solar phenomena.
- Built full-stack applications handling high-volume scientific data pipelines using Java, PostgreSQL, RabbitMQ, and web visualization frameworks.
- Contributed to nationally used platforms for GNSS-based ionospheric monitoring and space weather forecasting.

Tools for Open Multi-risk assessment using Earth Observation data, TOLOMEO

AFFILIATE RESEARCHER IN REMOTE SENSING

Università degli studi di Pavia, Pavia, Italy

Jul. 2013 - Jan. 2014

- **BREC Road Extractor Operator:** Development of roads network extractor in radar images, using C++ language within InterIMAGE platform (PUC/Rio)
- **BREC Road Cross Detection:** Development filters of polygons to detect intesection or cross paths of roads, using C++ language within InterIMAGE plataform (PUC/Rio)

Tools for Open Multi-risk assessment using Earth Observation data, TOLOMEO

AFFILIATE RESEARCHER IN REMOTE SENSING

Leibniz Universität Hannover, Hannover, Germany

Jan. 2013 - Jul. 2013

- **Network Snakes:** Study of a methodology to extract road centerlines based in a non-parametric algorithm called Active Contour Models (Snakes)

Extracurricular & Certifications

AWS Solution Architect - Associate - SAA-C03

CLOUD COMPUTING

Credly: <https://l1nq.com/tBam8>

Feb. 2025

Pattern Recognition and Computer Vision (Lecture)

STUTTGART UNIVERSITÄT

Stuttgart, Germany

Oct. 2017

FF-225 Laser I - Princípios Físicos (Lecture)

INSTITUTO TECNOLÓGICO DE AERONÁUTICA, ITA

SJC, São Paulo, Brazil

Ago. 2014

Publications

- Accurate mapping of Brazil nut trees using WorldView-3 and CNNs, Ecological Informatics, 2021
- 3D façade labeling using CNNs and SfM, Remote Sensing, 2018
- U-Net for forest mapping in high-resolution imagery, Remote Sensing in Ecology & Conservation, 2019

Full publication list available upon request or via [Google Scholar](#).